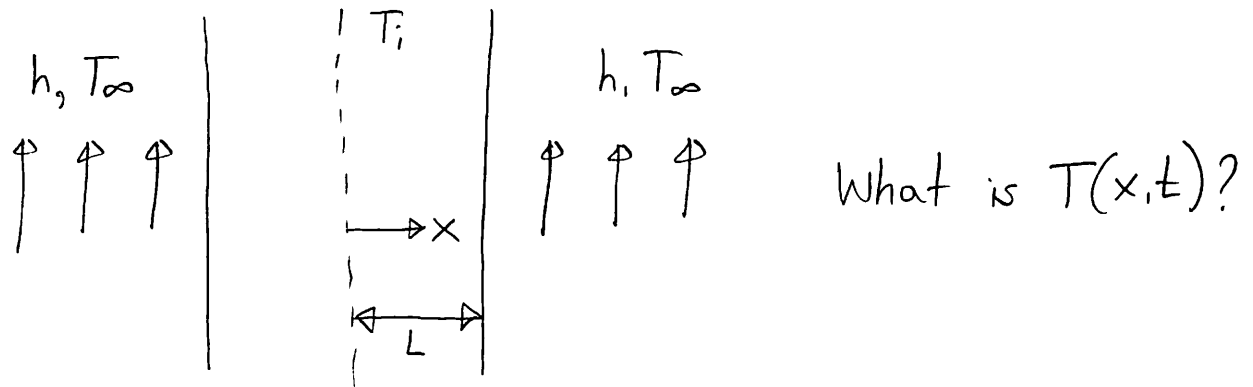


Transient Heat Conduction in Finite Bodies

So far, we've covered the cases of: 1) Lumped capacitance, $T \neq f(x)$
2) Infinite media, $0 \leq x \leq \infty$

How do we handle bodies that are finite in size & have $Bi_L > 0.1$?

Let's solve the plane wall problem. Slab initially placed in a medium with T_∞ & h . The slab has initial temp. T_i .



Looking at our heat equation:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{Q'''}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

Assuming: $\left. \begin{array}{l} 1) 1D \\ 2) Q''' = 0 \end{array} \right\} \frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$

We need 2 B.C.'s and an IC to solve:

1) $T(x, t=0) = T_i$ (Initial condition)

2) $\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0$ (1'st B.C.)

3) $-k \left. \frac{\partial T}{\partial x} \right|_{x=L} = h(T(x=L) - T_\infty)$ (2'nd B.C.)

To make our solution more tractable, we will non-dimensionalize

Let: $\bar{x} = \frac{x}{L}$ (Dimensionless distance from center, $0 \leq \bar{x} \leq 1$)

$$\frac{\partial \bar{x}}{\partial x} = \frac{1}{L} \Rightarrow \partial \bar{x} = \frac{\partial x}{L}$$

$$\Theta = \frac{T - T_{\infty}}{T_i - T_{\infty}} \quad (\text{Dimensionless temperature, } 0 \leq \Theta \leq 1)$$

$\circ (T_i, T_{\infty} = \text{constant})$

$$\frac{\partial \Theta}{\partial T} = \frac{\partial}{\partial T} \left(\frac{T}{T_i - T_{\infty}} \right) - \frac{\partial}{\partial T} \left(\frac{T_{\infty}}{T_i - T_{\infty}} \right) = \frac{1}{T_i - T_{\infty}} \Rightarrow \partial \Theta = \frac{\partial T}{T_i - T_{\infty}}$$

How about dimensionless time?

$$\frac{\partial T}{\partial x} = \frac{(T_i - T_{\infty})}{L} \frac{\partial \Theta}{\partial \bar{x}}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{T_i - T_{\infty}}{L^2} \frac{\partial^2 \Theta}{\partial \bar{x}^2}$$

$$\frac{\partial T}{\partial t} = (T_i - T_{\infty}) \frac{\partial \Theta}{\partial t}$$

} Back substituting into our PDE

$$\frac{T_i - T_{\infty}}{L^2} \frac{\partial^2 \Theta}{\partial \bar{x}^2} = (T_i - T_{\infty}) \frac{\partial \Theta}{\partial t} \cdot \frac{1}{\alpha}$$

$$\frac{\partial^2 \Theta}{\partial \bar{x}^2} = \frac{L^2}{\alpha} \frac{\partial \Theta}{\partial t} \quad \textcircled{1} \rightarrow \text{We still have a dimensional time here.}$$

Let's use Fo or dimensionless time (look on pg. 59 of notes)

$$Fo = \tau = \frac{\alpha t}{L^2} \equiv \frac{\text{diffusive transport rate}}{\text{storage rate}}$$

$$t = \frac{L^2 \tau}{\alpha} \Rightarrow \frac{\partial t}{\partial \tau} = \frac{L^2}{\alpha} \Rightarrow \partial t = \frac{L^2}{\alpha} \partial \tau \Rightarrow \text{back substitute into } \textcircled{1}$$

Now our equation becomes:

$$\frac{\partial^2 \theta}{\partial \bar{x}^2} = \cancel{\frac{L}{\cancel{x}}} \cdot \frac{\partial \theta}{\partial \tau} \cdot \cancel{\frac{\cancel{x}}{L}} \Rightarrow \boxed{\frac{\partial^2 \theta}{\partial \bar{x}^2} = \frac{\partial \theta}{\partial \tau} \text{ or } \frac{\partial \theta}{\partial Fo}}$$

↳ Dimensionless 1D, transient heat eq.

Now we can non-dimensionalize our IC & B.C.'s

1) Our IC becomes: $\boxed{\theta(\bar{x}, \tau=0) = 1}$

2) Our 1st B.C. becomes: $\boxed{\left. \frac{\partial \theta}{\partial \bar{x}} \right|_{\bar{x}=0} = 0}$

3) Our 2nd B.C. becomes:

$$-k \left. \frac{\partial T}{\partial x} \right|_{x=L} = h(T(x=L) - T_{\infty})$$

$$-k \frac{(\cancel{T_i} - T_{\infty})}{L} \left. \frac{\partial \theta}{\partial \bar{x}} \right|_{\bar{x}=1} = h \theta(\bar{x}=1) \cdot (\cancel{T_i} - T_{\infty})$$

$$\boxed{\left. \frac{\partial \theta}{\partial \bar{x}} \right|_{\bar{x}=1} = - \underbrace{\frac{hL}{k}}_{Bi} \theta(\bar{x}=1)}$$

$Bi \Rightarrow \text{Biot \#} \equiv \frac{\text{conduction resistance}}{\text{convection resistance}}$

Now we've reduced the problem to:

$$T(x, t) = f(x, L, t, \alpha, h, T_{\infty}, T_i)$$

⇓

$$\theta = f(\bar{x}, Bi, Fo) \Rightarrow \text{Very powerfull}$$

Now we can solve using the separation of variables:

$$\Theta(\bar{x}, \tau) = F(\bar{x}) \cdot G(\tau) \Rightarrow \text{Back substitute into our PDE and divide through by } F \cdot G$$

$$\underbrace{\frac{1}{F} \frac{\partial^2 F}{\partial \bar{x}^2}}_{f(\bar{x}) \text{ only}} = \underbrace{\frac{1}{G} \cdot \frac{\partial G}{\partial \tau}}_{f(\tau) \text{ only}} = \text{constant} \quad (\text{Must be a constant only since two independent functions})$$

Assuming our constant $= -\lambda^2$

$$\left. \begin{aligned} \frac{\partial^2 F}{\partial \bar{x}^2} + \lambda^2 F &= 0 \quad (1) \\ \frac{\partial G}{\partial \tau} + \lambda^2 G &= 0 \quad (2) \end{aligned} \right\} \text{Homogeneous ODE's. Use character. equation.}$$

Solving (1) first: $\lambda'^2 + \lambda^2 = 0$
 $\lambda' = \pm \sqrt{-1} \lambda = \lambda i$

So our solution becomes:

$$F = C_1 e^{+i\lambda \bar{x}} + C_2 e^{-i\lambda \bar{x}} \Rightarrow \text{since}$$

we can write our solution as:

$$F = C_1 \cos(\lambda \bar{x}) + C_2 \sin(\lambda \bar{x}) \quad (3)$$

Identities

$\sin \bar{x} = \frac{e^{i\bar{x}} - e^{-i\bar{x}}}{2i}$
$\cos \bar{x} = \frac{e^{i\bar{x}} + e^{-i\bar{x}}}{2}$

Now solving (2): $\lambda' + \lambda^2 = 0 \Rightarrow \lambda' = -\lambda^2$

$$G = C_3 e^{-\lambda^2 \tau} \quad (4)$$

Combining our two solutions, we obtain:

$$\Theta = F \cdot G = G_3 e^{-\lambda^2 \tau} (C_1 \cos(\lambda \bar{x}) + C_2 \sin(\lambda \bar{x}))$$

$$\Theta = e^{-\lambda^2 \tau} (A \cos(\lambda \bar{x}) + B \sin(\lambda \bar{x})) \quad (5)$$

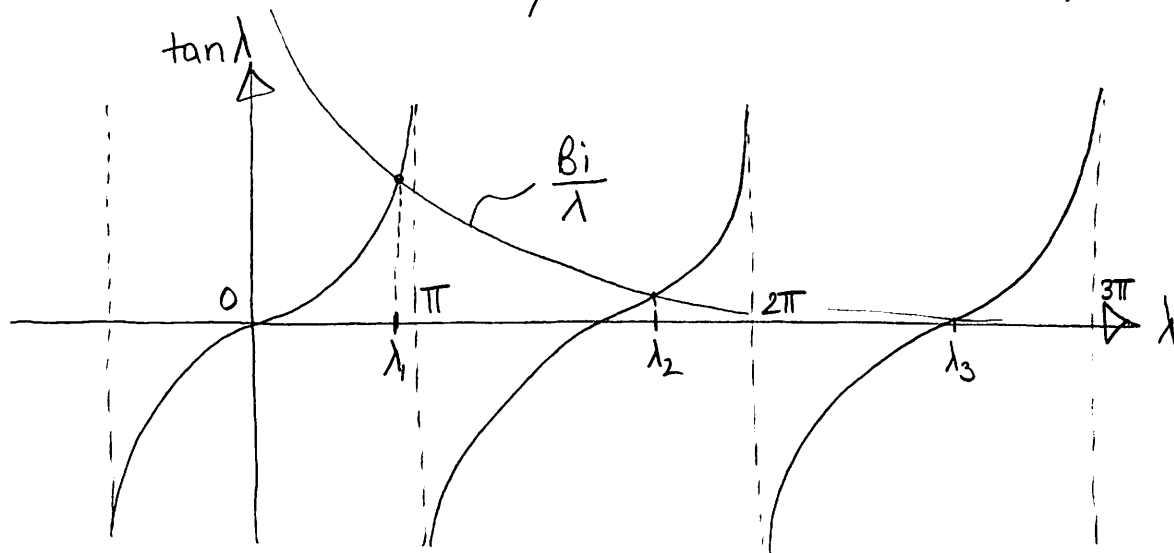
Now we apply our B.C.'s :

$$\left. \frac{\partial \Theta}{\partial \bar{x}} \right|_{\bar{x}=0} = 0 = -e^{-\lambda^2 \tau} (A \lambda \sin(0) + B \lambda \cos(0)) \Rightarrow \boxed{B=0}$$

$$\left. \frac{\partial \Theta}{\partial \bar{x}} \right|_{\bar{x}=1} = -Bi \Theta(\bar{x}=1, \tau) \Rightarrow -A e^{-\lambda^2 \tau} \lambda \sin \lambda = -Bi A e^{-\lambda^2 \tau} \cos \lambda$$

$$\lambda \tan \lambda = Bi \quad (6)$$

We know $\tan \lambda$ is a periodic function with period π , so solution can lie anywhere between $0 \leq \lambda \leq \pi$, $\pi \leq \lambda \leq 2\pi$, etc...



Multiple roots exist, so we have multiple solutions

$$\lambda_n \tan \lambda_n = Bi \quad (\text{Eigenfunction with eigenvalues } \lambda_n)$$

There exist an infinite number of solutions of the form $A e^{-\lambda^2 \tau} \cos(\lambda \bar{x})$. The final solution is a linear superposition of all of them:

$$\theta = \sum_{n=1}^{\infty} A_n e^{-\lambda_n^2 \tau} \cos(\lambda_n \bar{x}) \quad (7)$$

Our IC allows us to determine our constants A_n :

$$\theta(\bar{x}, \tau=0) = 1$$

$$1 = \sum_{n=1}^{\infty} A_n \cos(\lambda_n \bar{x})$$

Using orthogonality $\Rightarrow$ multiply both sides by $\cos(\lambda_m \bar{x})$ & integrate:

$$\int_0^1 \cos(\lambda_m \bar{x}) d\bar{x} = \sum_{n=1}^{\infty} A_n \underbrace{\int_0^1 \cos(\lambda_n \bar{x}) \cos(\lambda_m \bar{x}) d\bar{x}}_{=0 \text{ if } n \neq m}$$

So our solution becomes:

$$\begin{aligned} \underbrace{\int_0^1 \cos(\lambda_n \bar{x}) d\bar{x}}_{\sin(\lambda_n \bar{x}) \Big|_0^1} &= A_n \underbrace{\int_0^1 \cos^2(\lambda_n \bar{x}) d\bar{x}} \\ &= \sin(\lambda_n(1)) - \cancel{\sin(0)} \\ &= \sin(\lambda_n) \end{aligned}$$

$$\begin{aligned} \cos^2 x + \sin^2 x &= 1 \\ \cos^2 x - \sin^2 x &= \cos 2x \\ \cos^2 x &= (1 + \cos 2x)/2 \\ \Rightarrow [2\lambda_n \bar{x} + \sin(2\lambda_n \bar{x})] \Big|_0^1 \\ &= 2\lambda_n + \sin(2\lambda_n) \end{aligned}$$

$$A_n = \frac{4 \sin \lambda_n}{2\lambda_n + \sin(2\lambda_n)} \quad (8) \Rightarrow \text{Combine with eq. 7 \& you're done!}$$